3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 1 – Introduction to Future Engineers & Technology

Complete Theory & Practical Notes

Topic :- Welcome to Future Tech Program

Learning Objectives:-

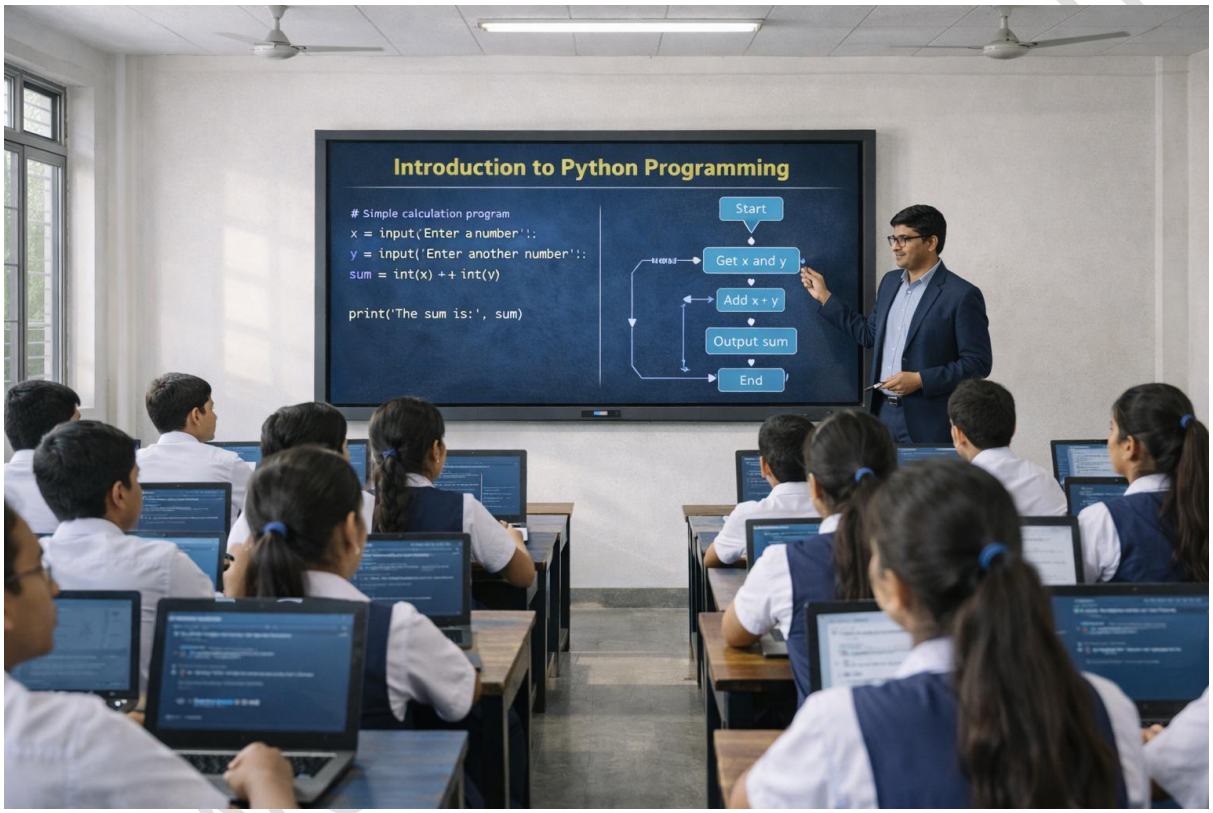
At the end of this class, students will be able to:

- Explain what engineering and technology mean.
- Identify different areas of modern technology.
- Understand why programming is important for engineers.
- Explain what Python is and where it is used.
- Open a Python programming environment.
- Write and execute their first Python program.
- Use print() to display information.
- Create a simple "Future Engineer" introduction program.

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DATA




Introduction to Python Programming
Simple calculation program
x = input('Enter a number':
y = input('Enter another number':
sum = int(x) + int(y)
print('The sum is:', sum)
Start
Get x and y
Add x + y
Output sum
End



Session Plan

Session	*Type*
Welcome & Icebreaker	Activity
Who is a Future Engineer?	Theory
Engineering & Technology Around Us	Theory + Discussion
Where Does Python Fit?	Theory
Python Setup & First Program	Practical
Break / Quick Challenge	Activity
Python Basics: `print()`	Practical
Mini Project: Future Engineer Profile	Practical
min Quiz + Review + Challenge	Assessment

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1. WELCOME & ICEBREAKER

Teacher Introduction:-

Welcome, Future Engineers!

Today we are starting a journey where you will learn how to use technology to solve problems and create new things.

What You Will Learn

What is Future Engineers?

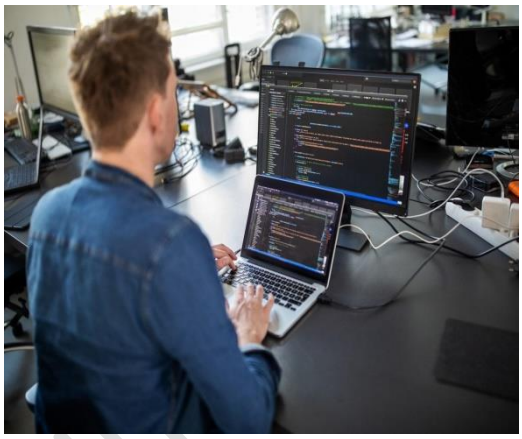
Future Engineers is a technology education program that prepares students for the future world of Artificial Intelligence, Programming, Robotics, Data Science, and Cybersecurity.

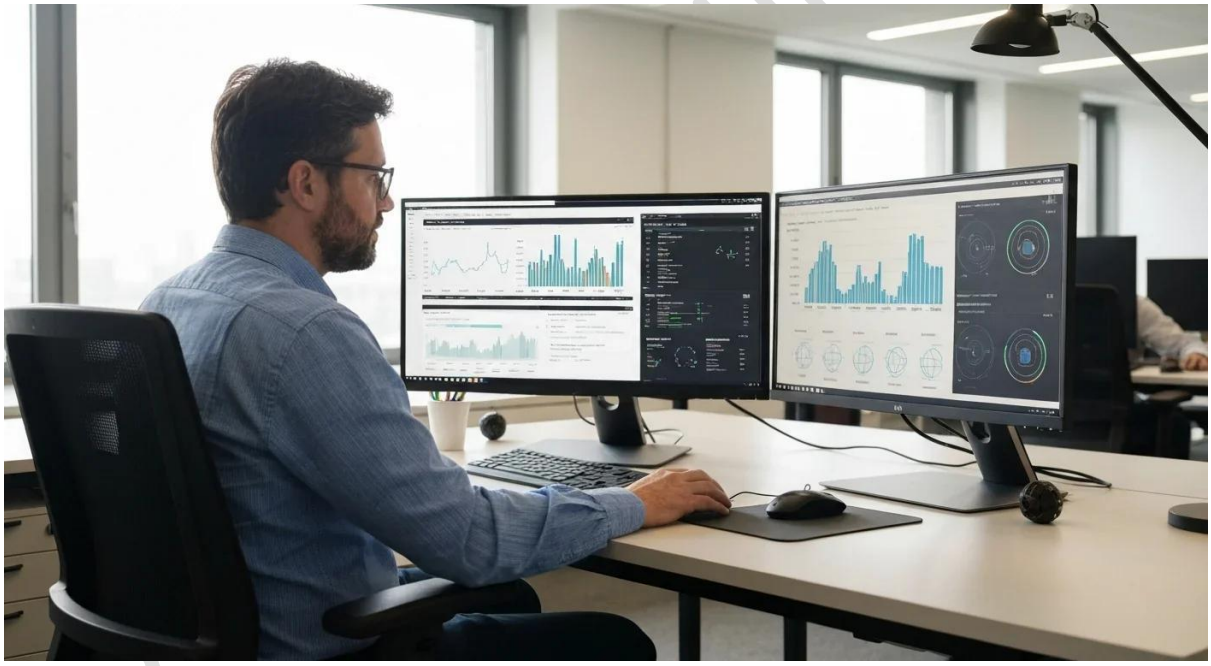
- It teaches students how to:

- Solve real-life problems.
- Create software applications.
- Understand Artificial Intelligence.
- Build innovative technology projects.
- Become future innovators and engineers.

What Does a Future Engineer Do?







2. WHO IS A FUTURE ENGINEER?

An engineer is a person who uses:

Science + Mathematics + Creativity + Technology + Problem Solving

to design and build solutions.







Engineer	What they work with
Software Engineer	Programs and applications
Mechanical Engineer	Machines and mechanical systems
Civil Engineer	Buildings, roads, bridges
Robotics Engineer	Robots and automation
AI Engineer	Artificial intelligence
Aerospace Engineer	Aircraft and spacecraft
Biomedical Engineer	Healthcare technology

3 . ENGINEERING & TECHNOLOGY AROUND US

Introduce students to major technology areas.

1. Artificial Intelligence — AI

Computers can learn patterns and make decisions.

Examples:

- Voice assistants
- Chatbots
- Image recognition
- Recommendation systems

2. Robotics

Robots combine:

Mechanical parts + Electronics + Sensors + Programming

3. Internet of Things — IoT

Everyday objects can connect to the internet.

Examples:

- Smart watch
- Smart bulb
- Smart door
- Smart agriculture

4. Automation

A machine performs a task automatically.

Example:

Sensor → Computer → Decision → Motor

5. Space Technology

Examples:

- Satellites
- Rockets
- Space probes
- Mars rovers

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6. Software Technology

Examples:

- Websites
- Mobile apps
- Games
- Operating systems
- AI applications







4 .WHERE DOES PYTHON FIT?

What is programming?

Simple explanation:

Programming is giving instructions to a computer so that it can perform a task.

For example:

Human:

Calculate 10 + 20.

Computer needs instructions to perform that task.

Python:

```
print (10 + 20)
```

Output:

```
30
```

WHAT IS PYTHON?

Explain:

Python is a programming language used to give instructions to computers.

Python is popular because its syntax is relatively easy for beginners to read and write.

Python is used in:

- Robotics
- Artificial Intelligence
- Data Science
- Web Development
- Games
- Scientific Computing
- Automation
- Engineering

5 .FIRST PYTHON PRACTICAL

Students now move to computers.

They can use any beginner-friendly Python environment available at your institution.

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First Program

Ask students to type:

```
print("Hello, Future Engineer!")
```

Run the program.

Expected output:

```
Hello, Future Engineer!
```

Explain

`print()` means:

Display something on the screen.

The text inside quotation marks is called a string.

QUICK CHALLENGE

Give students a short break.

Then ask:

Challenge

Change the program to display:

```
My name is _____.
I am a Future Engineer.
I love Technology.
```

Example:

```
print("My name is Rahul.")
print("I am a Future Engineer.")
print("I love Technology.")
```

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6 . *PYTHON PRACTICAL*

Activity 1 — Print Different Messages

Students try:

```
print("Welcome to Python")
print("I want to become an engineer")
print("I can solve problems")
```

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Activity 2 — Numbers

```
print(10)
print(25)
print(100)
```

Then:

```
print(10 + 20)
print(50 - 10)
print(5 * 5)
```

Expected:

```
30
40
25
```

Explain that Python can perform calculations.

Teacher Demonstration — Computer as a Calculator

Ask students:

Question:

What is:

25 + 75?

Students guess.

Then run:

```
print(25 + 75)
```

Python gives:

```
100
```

Now:

```
print(12 * 8)
```

Output:

```
96
```

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7 .MINI PROJECT

PROJECT: "MY FUTURE ENGINEER PROFILE"

Students create their first mini Python project.

Starter version

```
print("===== MY FUTURE ENGINEER PROFILE =====")
print("My name is Alex")
print("I am learning Python")
print("My favorite technology is Robotics")
print("I want to become an AI Engineer")
print("I will solve problems using technology")
```

Students replace the information with their own.

Level 2 Challenge

Add calculations:

```
print("===== MY FUTURE ENGINEER PROFILE =====")

print("My name is Alex")
print("I am 12 years old")

print("I have been learning Python for")
print(2 + 1)

print("My favorite technology is Robotics")
print("My dream is to become an Engineer")
```

BONUS CHALLENGE

Ask advanced students to create a Future Engineer Robot Message**:**

```
print("□ ROBOT ONLINE")
print("Hello Future Engineer!")
print("My name is Robo-1")
print("I am ready to learn Python")
print("Mission: Solve Problems")
```

What instructions would you give your robot?

Students can invent commands such as:

- Move forward
- Turn left
- Turn right
- Pick up object

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- Stop

This creates a bridge from **Python → Robotics → Engineering**.

8 . REVIEW & QUIZ

Quick Quiz

1. What is an engineer?

A. Someone who only repairs machines
B. Someone who uses knowledge to design and solve problems
C. Someone who only uses computers

Answer: B

2. What is Python?

A. A programming language
B. A robot
C. An operating system

Answer: A

3. What does `print()` do?

A. Deletes the computer
B. Displays information
C. Turns off Python

Answer: B

4. What will this produce?

```
print(10 + 5)
```

Answer: 15

5. Name two technologies used by engineers.

Possible answers:

- AI
- Robotics
- IoT
- Computers
- Sensors
- Automation

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